Midterm Exam Sample

1. **Multiple Choice or Multiple Answers**
   For questions with square options (☐), select all that apply.
   (a) Assume x and y get multiplied together element-wise: $x * y$. Let S_x and S_y be the shapes of x and y respectively. For which of the following will Numpy or Tensorflow throw a shape error?
   - ◯ $S_x = [100, 10], S_y = [10, 100]$
   - ◯ $S_x = [1, 10], S_y = [10, 10]$
   - ◯ $S_x = [1, 10], S_y = [1, 10]$
   - ◯ $S_x = [1, 10, 100], S_y = [1, 1, 1]$

   (b) When the gradient flows backward through the sigmoid activation functions in backpropagation, it cannot change sign.
   - ◯ True
   - ◯ False

   (c) Which of the following statements is true about skip-gram?
   - ☐ It predicts the center word from the surrounding context words
   - ☐ The final word vector for a word is the average or sum of the center vector **v** and context vector **u** corresponding to that word
   - ☐ On small corpora, it has better performance than Glove
   - ☐ It makes use of global co-occurrence statistics

2. **Gradient Descent** Adagrad is a modification to the stochastic gradient descent algorithm used to update parameters. Updates are performed using the following formula:

$$c^{(t)} = \sqrt{(g^{(1)})^2 +, ..., +(g^{(t-1)})^2 + \epsilon}$$

$$\theta_{t+1} = \theta_t - \frac{\eta}{c^{(t)}} g^{(t)}$$

   where θ is the parameter to be updated, $g^{(t)}$ is the gradient of the loss with respect to θ, ϵ is a smoothing parameter and η is the learning rate. $c^{(0)}$ is initialized to zero at the start of the algorithm. State two differences between Adagrad and SGD in terms of step size (learning rate) and describe what effect they would have. Note: This is more of a machine learning question.

3. **Advanced RNNs**
   (a) Assume you choose to use LSTM instead of a vanilla RNN in a language modeling task. In one LSTM cell, how many gates are defined and what are they?

   (b) In your LSTM model, if the input vector: $x_t = 0$, then you get: $h_t = h_{t-1}$.
   - ◯ True
   - ◯ False

   (c) The outputs of all the gates in the LSTM model are non-negative values.
   - ◯ True
   - ◯ False